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Key Points:

- We measure the relief of 672 and 584 shortening structures in Mercury's smooth plains and intercrater plains units, respectively
- Areal strain estimates for the smooth plains and intercrater plains are similar, suggesting both units record similar amounts of shortening strain from global contraction
- Differences in morphology and fault density of shortening structures may reflect differences in lithospheric structure at fault initiation

Supporting Information:

- Supporting Information S1
- Figure S1
- Figure S2
- Data Set S1
- Table S1
- Table S2

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Distribution of Areal Strain on Mercury: Insights Into the Interaction of Volcanism and Global Contraction

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Abstract Shortening tectonic structures within Mercury's two largest geological units display a clear contrast in relief, length, and spatial density. The volcanic smooth plains units are deformed by smaller-scale structures yet host more features per area than the older intercrater plains. Although faulting in the intercrater plains is dominantly attributed to global contraction, it has been unclear whether the smooth plains faults result from volcanic loading, global contraction, or both. We use estimates of fault length and displacement to calculate spatial variations of areal strain for each unit. We find that high strain concentrations within the smooth plains suggest that global contraction has contributed to deformation in these units. The observed contrast in morphology and spatial density of structures between units may primarily reflect differences in mechanical and/or structural characteristics of the lithosphere when faulting was initiated.

Plain Language Summary Slow cooling of Mercury's mantle and core has decreased the radius of the planet by as much as 7 km. This radial contraction put the crust in a state of horizontal compression resulting in widespread thrust faulting. Extensive flood volcanism resurfaced portions of the crust, the most recent deposits of which are termed producing the *smooth plains*. Thrust faults are also observed within these plains, but it is not clear whether these features formed from global contraction or from bending of the crust due to the weight of the emplaced lavas. In this study, we measure the length of the faults and their relief (height) to calculate the spatial distribution of strain (horizontal shortening) in both the smooth plains and intercrater plains of Mercury. We find that the amount of strain recorded in the smooth plains is best explained by at least some contribution from stresses due to global contraction. Therefore, we suggest that all geological units on Mercury have been deformed by global contraction, indicating that substantial radius decrease occurred after the bulk of volcanic activity had ended.

1. Introduction

The surface of Mercury is dominated by thousands of tectonic landforms interpreted to be the expression of thrust fault-related folding (Byrne et al., 2014; Strom et al., 1975; Watters et al., 2004, 2015; Watters & Nimmo, 2010), resulting from secular cooling and contraction of Mercury's interior (Tosi et al., 2013; Solomon, 1977). In particular, tectonic structures in the planet's dominant physiographic terrain, the *intercrater plains* (ICP) units, are thought to reflect substantial crustal shortening from global contraction, suggesting a cumulative decrease in the planet's radius of up to 7 km (Strom et al., 1975; Solomon, 1977; Byrne et al., 2014). The ICP are the oldest and most extensive geological terrain on Mercury, with no evidence for major resurfacing since ~3.8 Ga (Byrne et al., 2016; Whitten et al., 2014).

In the planet's smooth plains (SP) units, shortening structures typically have less relief and are shorter than those in the ICP. The SP show evidence of flood volcanism that is young relative to the ICP but that mostly ceased by around 3.5 Ga (Byrne et al., 2016; Denevi et al., 2013). Borealis Planitia, also termed the *northern smooth plains* (NSP), is the largest such deposit (Figure 1). Shortening structures found in the SP are regarded as analogous to the wrinkle ridges in the lunar maria (Melosh, 1978) and may reflect, at least in part, loading-induced stresses from the volcanic infill (Byrne et al., 2014; Melosh & McKinnon, 1988; Solomon & Head, 1979; Watters et al., 2015). Furthermore, the style of faulting in the SP may have been influenced by mechanical detachments within volcanic strata producing "thin-skinned" faulting (i.e., faulting restricted mainly to the volcanic units), whereas ICP shortening structures may be "thick-skinned" (i.e.,

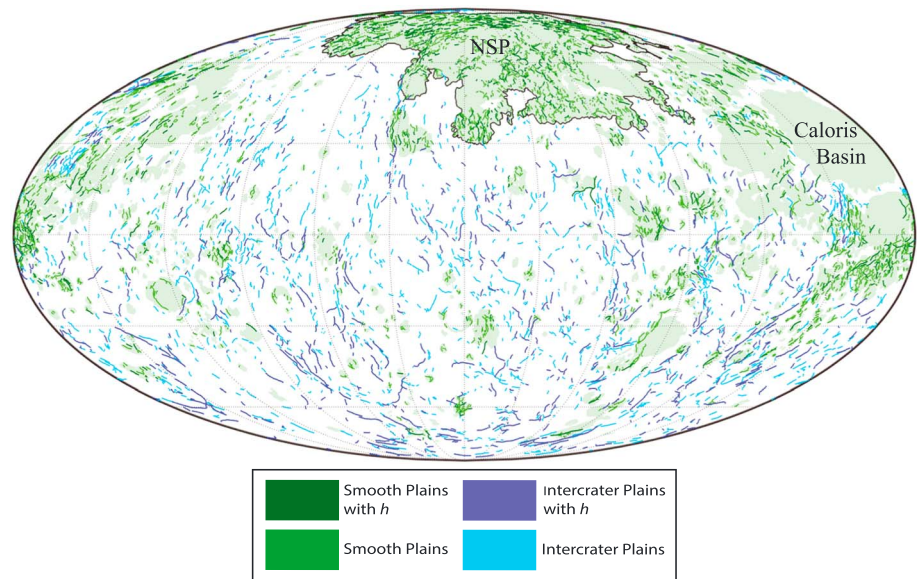


Figure 1. Global distribution of shortening features (Byrne et al., 2014) and the SP (green) and ICP (white) units. Locations of the 672 SP and 584 ICP faults surveyed in this study are shown in dark-green and blue, respectively. The locations of the Caloris basin and the NSP are labeled. NSP = northern smooth plains; SP = smooth plains; ICP = intercrater plains.

deeper faulting that may even have penetrated to the contemporary brittle-ductile transition) (Byrne et al., 2014; Watters et al., 2004; Watters et al., 2015). Together, these characteristics suggest that the total shortening strain, and in particular that due to global contraction, should be less in the SP than the ICP. However, the global distribution of tectonic features mapped using data from the MErcury Surface, Space ENvironment, GEochemistry, and Ranging (MESSENGER) mission has revealed that the SP host a disproportionately large number of faults per area compared with the ICP (Byrne et al., 2014). The SP comprise ~27% of the total planetary surface, yet host ~63% of the mapped shortening structures and ~50% of the total mapped fault length (Byrne et al., 2014).

In this study, we analyzed the geographical distribution of strain recorded by shortening structures in order to investigate the strain accommodated by all SP units collectively, as well as by the NSP alone. We compared these results with the strain hosted by the ICP. If the SP and/or NSP hold at least as much areal strain as the ICP, this suggests that the SP/NSP have accommodated more deformation than previously recognized. Thus, not only the observed geometries and morphologies but also the spatial densities of the ICP and SP fault populations can elucidate the dominant process(es) responsible for faulting in these units. Additionally, determining the tectonic process(es) responsible for faulting in the SP has implications for the timing and duration of global contraction and the thermal evolution of the planet. Stratigraphic relationships of shortening structures with volcanically buried impact craters show that crustal shortening was prolonged and continued after SP emplacement (Banks et al., 2015), providing an observational basis for deformation of these units by global contraction.

2. Characterizing Fault Morphology

Commonly, ICP shortening structures are classified from cross sectional shape as either “lobate scarps,” which are asymmetric with one steep side, or “high-relief ridges,” which exhibit a symmetrical arch (Dzurisin, 1978; Melosh & McKinnon, 1988; Strom et al., 1975). Within the SP, shortening structures are generally referred to as *wrinkle ridges*, and in cross section have broad, low-relief arches (sometimes with a superposed crenulation). However, some SP structures have a lobate scarp morphology (Figure 2; Byrne et al., 2014), and some ICP structures show an arch-like morphology.

We used contractional features mapped by Byrne et al. (2014), combined with topography from the Mercury Laser Altimeter in the northern hemisphere (Zuber et al., 2012) and from stereo derived topography in the

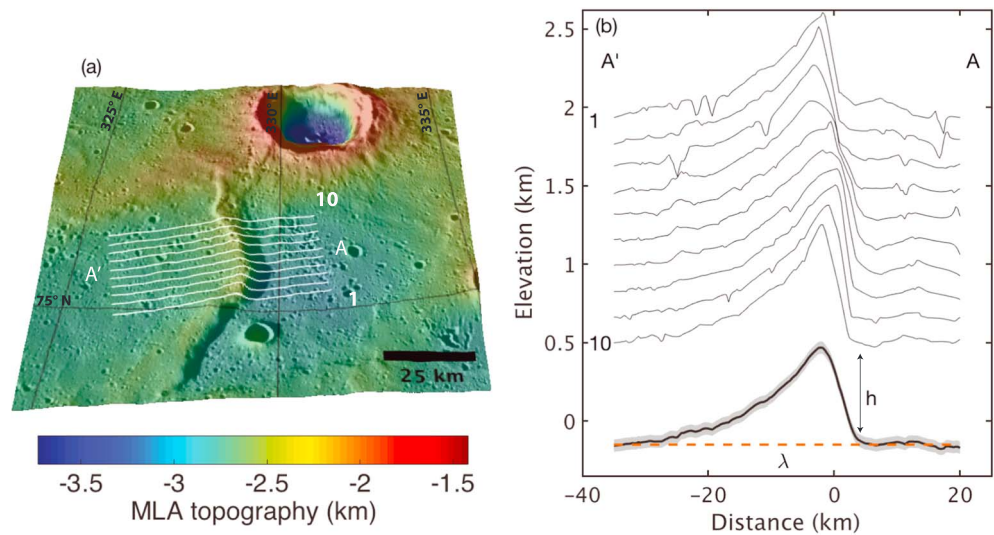


Figure 2. Example of relief (h) and width (λ) estimates for a shortening structure with an asymmetric cross section in the smooth plains. (a) MLA topography, with profiles 1 to 10 taken across the landform. (b) Individual profiles for tracks 1 to 10, offset vertically for clarity. The average detrended topography (black) and ± 1 standard deviation (gray shaded region) are shown below the individual profiles. Fault relief above the baseline (red dashed), and fault width are labeled. MLA = Mercury Laser Altimeter.

southern hemisphere (Becker et al., 2016), to systematically survey and characterize fault-related morphology. The spatial resolution of the topography data ranges from ~ 250 to 665 m per pixel. Within the Caloris basin, Mercury's largest impact basin, shortening structures were excluded and the corresponding surface area was subtracted from strain calculations (cf. Byrne et al., 2014). This basin hosts a complex structure of overlapping tectonic structures, and the extent of the role played by basin-related tectonics is unclear (Melosh and McKinnon, 1988; Cunje & Ghent, 2016; Klimczak et al., 2010; Kennedy et al., 2008).

The relief (h) of shortening structures was measured by extracting 10 topographic profiles perpendicular to strike across each feature (Figure 2). Each profile was detrended, and the average and standard deviation of the 10 profiles taken. Each average profile (and its standard deviation) was used to visually estimate a background “zero” level baseline (Figure 2) and to calculate the relief, h . The uncertainty (σ) in h was determined from the average standard deviation of the topography, which for a given thrust fault was typically about ± 90 m. The width (λ) of the structure was recorded, as was the basic shape (asymmetric or symmetric).

Relief and width estimates were obtained for 584 structures in the ICP and 672 in the SP, representing 26% and 19% of the populations of the shortening structures therein, respectively (Byrne et al., 2014). Although the map results from Byrne et al. (2014) informed the choice of landforms analyzed here, our morphometric measurements are new. The remaining faults could not be characterized robustly because of limitations in the spatial resolution of topographic data, the presence of cross-cutting structures, and/or superposed crater material.

In the SP, h varies between 0.06 and 1.7 km, with a median of 0.31 km. In contrast, within the ICP, h varies between 0.09 and 3 km, with a median of 0.57 km. This difference in relief is consistent with previous studies (Byrne et al., 2014; Watters et al., 2015; Melosh & McKinnon, 1988; Watters & Nimmo, 2010; Strom et al., 1975). In the SP, $\sim 80\%$ of structures have h less than 0.5 km, compared with only $\sim 40\%$ of structures in the ICP (Figure S1 in the supporting information). However, the statistical distributions of λ for shortening structures in the SP and ICP are comparable, with a median λ of 33 and 37 km, respectively.

3. Strain Estimates

3.1. Approach

The calculation of strain requires knowledge of the displacement (D) along individual fault planes, which can be obtained from the relationship $D = h/\sin(\theta)$, where θ is the fault plane dip angle. The horizontal

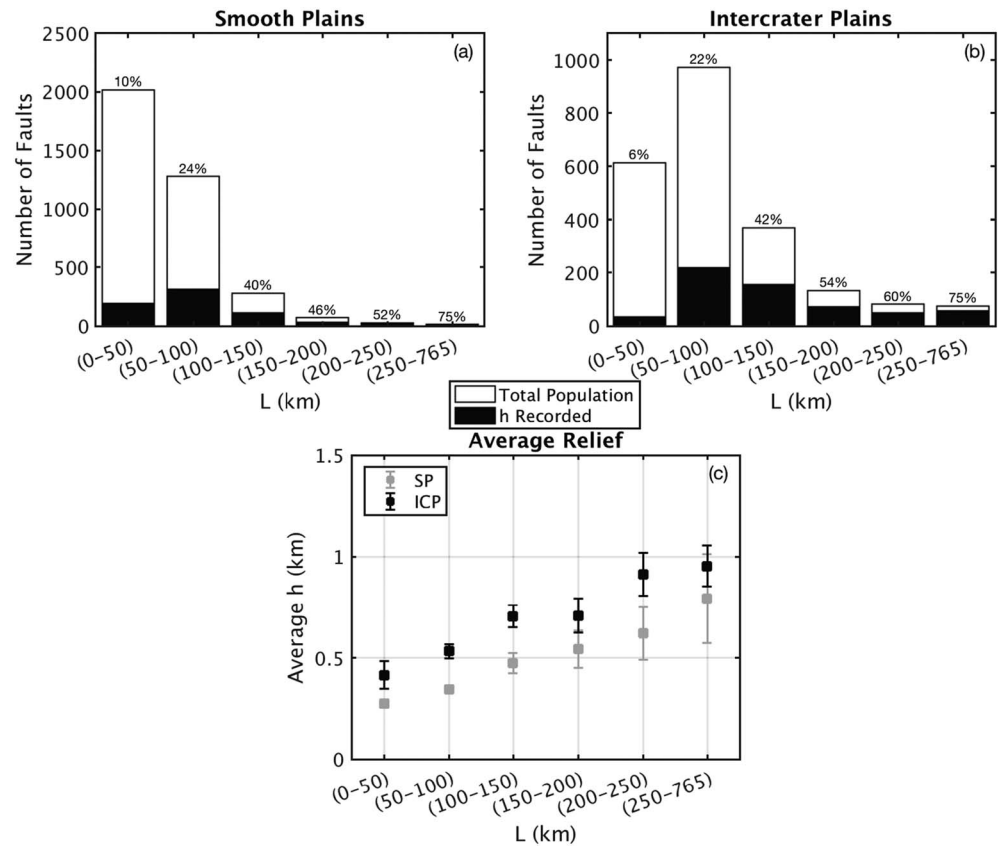


Figure 3. Histograms of fault lengths for (a) the SP and (b) the ICP, showing the total fault population (black plus white) compared to the number of shortening structures for which h was used to calculate D (black). The percentage of shortening structures with h estimates (see section 2) in each bin is also given. (c) Mean relief for various length bins (gray, SP; ICP, black). SP = smooth plains; ICP = inter crater plains.

shortening accommodated by a single fault can be characterized by $D L \cos(\theta)$ (Byrne et al., 2014; Scholz & Cowie, 1990). The regional areal strain, represented by a population of N faults in an area A is given by

$$\epsilon = \frac{1}{A} \sum_{k=1}^N D_k \cos(\theta) L_k \quad (1)$$

To enable a direct comparison with the previous literature for Mercury, θ values of 25° , 30° , and 35° were used (Byrne et al., 2014; Strom et al., 1975; Watters & Nimmo, 2010).

For the population of faults for which h , and thus the horizontal shortening, could not be robustly characterized, an h value was estimated and assigned to an individual fault based on its mapped L (Byrne et al., 2014). Figures 3a and 3b show the number of faults (black) for which h estimates were used to obtain D (binned into L ranges) compared to the number of faults that did not have such estimates (white). For structures with measurable relief, we then computed the mean h and the 95% confidence limits in each bin (Figure 3c). These values were used to assign an h value (and a corresponding uncertainty) to each fault without a measured relief depending on its L bin. D was then estimated for a given θ for every fault, and the strain calculated using equation (1).

The mean h in each bin increases as L increases in both the ICP and the SP (Figure 3c) and is well constrained because of the substantial number of faults with h estimates (Rice, 1995). Furthermore, features with L greater than 100 km were the best sampled, which is important as the longest length faults hold the most strain (see section 4.1).

Table 1*For a Given θ , the Strain for the SP, the NSP Alone, the ICP, and the Global Estimate Were Calculated Using Equation (1)*

	Number of structures	$\theta = 25^\circ$	$\theta = 30^\circ$	$\theta = 35^\circ$
SP	3,670	0.0090 ± 0.0008	0.0072 ± 0.0006	0.0060 ± 0.0005
NSP	1,675	0.0082 ± 0.0007	0.0067 ± 0.0006	0.0054 ± 0.0005
ICP	2,261	0.0054 ± 0.0005	0.0043 ± 0.0004	0.0036 ± 0.0003
Global	5,934	0.0063 ± 0.0005	0.0051 ± 0.0004	0.0042 ± 0.0004
Amount of global contraction				
Radius change (this study)	5,934	7.7 ± 0.6	6.2 ± 0.5	5.1 ± 0.5
Radius change (Byrne et al., 2014)	5,934	7.1	5.7	4.7

Note. Areas of 1.9×10^7 , 5.8×10^6 , and 5.4×10^7 km² were used for the SP, NSP, and ICP, respectively (Denevi et al., 2013). The inferred planetary radius change was calculated from the global strain estimate for all the structures and compared to radius change estimates reported by Byrne et al. (2014). NSP = northern smooth plains; SP = smooth plains; ICP = intercrater plains.

3.2. Results

We find that the SP, and specifically the NSP, have accommodated more shortening strain than the ICP (Table 1). Even though, on average, thrust faults in the ICP have greater relief and are longer, the higher spatial density of SP thrust faults means that the areal strain hosted by the ICP is not more than that hosted by the SP. This interpretation assumes that thrust faults in the ICP and SP have the same average dip angle, but fault dips are poorly known. However, even if the average dip in the SP and ICP is 35° and 25° , respectively, our calculations show that the SP still hosts more strain.

Furthermore, a global strain estimate using all the structures in both units was used to estimate the decrease in Mercury's radius, ΔR , under the assumption that most of the faulting results from global contraction. The resulting ΔR for a given θ agrees, within uncertainties, with the estimates of Byrne et al. (2014) provided in Table 1. Byrne et al. (2014) estimated global contraction following the procedure used by Scholz and Cowie (1990). This approach fits a linear regression model to the displacement-length data. Fault growth models predict that with increased displacement, fault length should increase linearly (Scholz & Cowie, 1990). To compare our measurement results to these studies, we also applied this approach in section S2. Although this method produces very similar strain values for the ICP and SP (Table S1) to those given in Table 1 and to those obtained by Byrne et al. (2014), the scatter in the displacement-length plot is not being captured by a single linear regression fit.

4. Discussion

4.1. Strain Estimates and Implications for Timing of Global Contraction

The inferred areal strains (Table 1) suggest that the SP have accommodated at least as much, and possibly more, shortening strain than the ICP, in contrast to previous interpretations (Byrne et al., 2014; Melosh & McKinnon, 1988; Watters et al., 2015). However, strain estimates depend on the resolution of the data used to identify and map faults. The smallest faults surveyed in the ICP and the SP have lengths of ~ 10 km, but a population of smaller thrust faults (those with lengths less than 10 km and reliefs of ~ 15 –55 m) have been recently identified (Watters et al., 2016). We therefore investigated whether a large number of smaller faults could hide substantial strain. The number, N_L , of SP or ICP faults of length greater than length, L , can be described by the following function (Cowie et al., 1993):

$$N_L = N_t [\exp(-\mu L)] \quad (2)$$

where N_t is the total number of faults and μ is a fitted parameter. We used the distribution of mapped fault lengths to estimate the number of faults below the minimum fault length studied here (Figure 4a) and the strain that could be hosted by them (Figure 4b), following the approach of Cowie et al. (1993). The best fit parameters of equation (2) for the SP and ICP (Figure 4a) predict that 1,616 SP and 962 ICP faults could have lengths less than 10 km, which would increase the number of faults in each population by 44% and 42%, respectively. The μ determined for each fault population can be used to estimate the continuous probability density function, $F(L) = \mu \exp(-\mu L)$, and the expected value of strain contributed by faults with lengths less than 10 km is given by a continuum form of equation (3):

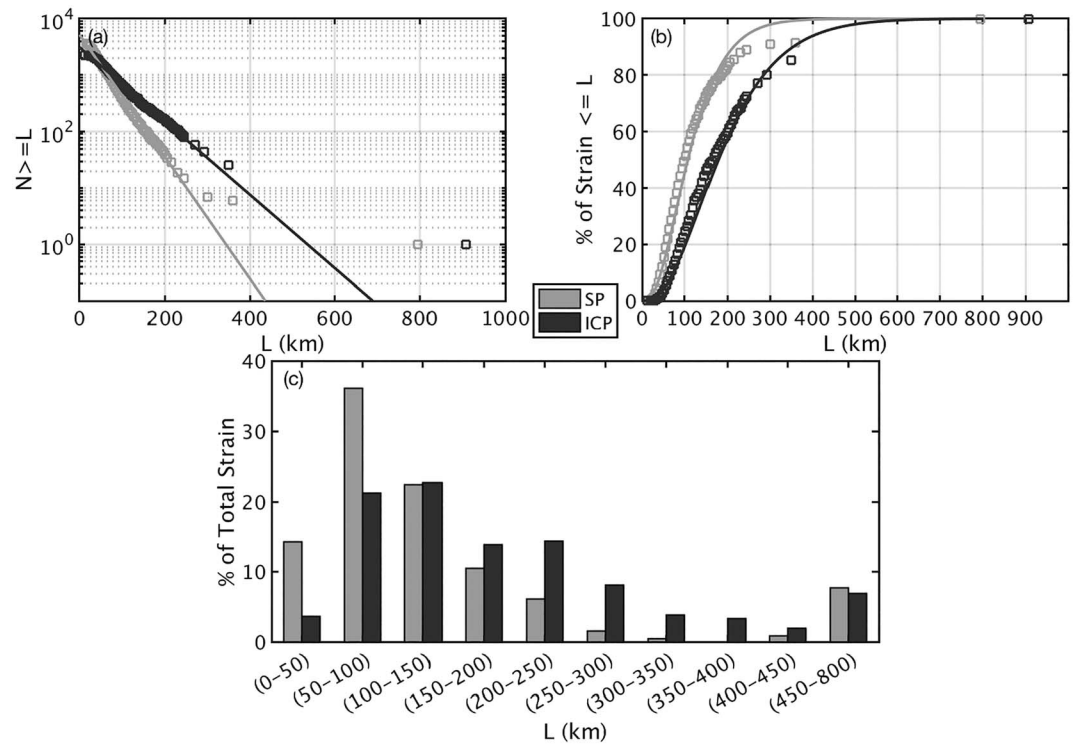


Figure 4. (a) Fault count cumulative distribution functions for the smooth plains (gray) and the intercrater plains (black) with best fit empirical forms (equation (2)) given by the solid lines. (b) Fault length cumulative distribution function normalized to 100%. Symbols and empirical fits as in (a). (c) Histograms of strain binned by fault length.

$$\epsilon = \frac{DN_L \cos \theta}{A} \int_0^{10} F(L) L dL \quad (3)$$

The resulting strain cumulative distribution functions (Figure 4b) demonstrate that almost 100% of the total strain is represented by faults with lengths greater than 10–20 km. Thus, even if a large number of faults shorter than 10 km exist on Mercury, they do not contribute significant strain in either the SP or ICP units. Furthermore, Figure 4b shows 40% and 60% of the strain in the ICP and SP is hosted by faults with $L > 100$ km, respectively. The majority of faults in this population had h characterized (i.e., we did not assume an h from Figure 3c). Therefore, strain estimates for the most important part of the population (largest L) are well characterized.

We suggest that the larger amount of strain in the SP is consistent with a substantial amount of global contraction occurring after the major episode(s) of plains volcanism that ended by about 3.5 Ga (Byrne et al., 2016; Denevi et al., 2013). This finding is consistent with observations of cross-cutting relationships of faults and craters (Banks et al., 2015; Crane & Klimczak, 2017) as well as small-scale faulting (Watters et al., 2015), and with thermal evolution models (Hauck et al., 2004; Tosi et al., 2013)—all of which suggest that the initiation of global contraction was contemporaneous with late-stage plains emplacement but continued into the geologically recent (Banks et al., 2015; Watters et al., 2016).

4.2. SP Versus ICP Shortening Structures

Figures 4a and 3c also document the contrast between the lengths and reliefs of mapped shortening structures in the ICP and SP. The ICP hosts approximately 58% more faults with lengths greater than 100 km than the SP (Figure S1) and has consistently larger h for all L . Nonetheless, the larger number of short faults in the SP means that the cumulative fault lengths are comparable: 2.08×10^5 and 2.06×10^5 km in the SP and ICP, respectively. Both the ICP and SP have substantial numbers of faults shorter than 50 km (40% in the ICP and 60% in the SP).

We suggest that the morphological differences between structures in the two units may not reflect the amount of global contraction-induced strain accommodated, but rather differences in how that strain was manifest. Variations in rheology and/or the structural character (e.g., fracturing) of the lithosphere could lead to strain that is more evenly distributed within the SP but concentrated into fewer, larger structures in the ICP (e.g., Byrne et al., 2014). Layers of mechanical weakness (acting as detachment surfaces or décollements) within the SP lavas (cf. Byrne et al., 2014; Mueller & Golombek, 2004) could promote the formation of many small landforms for a given strain. Furthermore, field data on Earth and numerical modeling suggest that fault spacing is related to the mechanical thickness for confined fault populations controlled by buckling/localization instabilities (Montési & Zuber, 2003; Zuber & Aist, 1990) or shear lag (Banerdt & Sammis, 1992). If, because of elevated heat flow, the SP had a thin elastic lithosphere at the time of fault initiation, shortening could have led to more closely spaced faults, a result predicted by fault localization models for Mercury (Zuber et al., 2010). This scenario requires that a substantial amount of global contraction occurred after the major volcanic units were emplaced, consistent with observations (e.g., Crane & Klimczak, 2017) and current thermal models (Tosi et al., 2013).

Elastic dislocation modeling of SP and NSP shortening structures require thrust faults that penetrate ~15 km into the lithosphere (Peterson et al., 2017), suggesting that at least a subpopulation of SP faults are not confined to the volcanic strata. Head et al., 2008 estimated the thickness of the volcanic plains to be within ~3 km from infilled craters. Furthermore, although faults within the SP and NSP may extend to greater depths than previously thought (Byrne et al., 2014; Watters et al., 2004; Watters et al., 2015), on average the depth of faulting was less than for the ICP shortening structures (Peterson et al., 2017). This finding, in turn, supports our inference that the thickness of the mechanical layer played a controlling role in fault spacing across Mercury. Simply put, if accommodation of global contraction is relatively limited in the vertical direction, then fault spacing will decrease to absorb a given amount of contraction.

The higher strain inferred for the SP and NSP compared to that for the ICP, as well as the increased spatial density of structures in the SP, could reflect additional strain from load-induced stresses from the volcanic infill that resurfaced the terrain. Faulting induced by bending stresses from volcanic loading would likely be confined to the volcanic layer, that is, be “thin-skinned” (Byrne et al., 2014; Watters et al., 2015). If wrinkle ridges in the SP penetrate only to the depth of the volcanic cover rather than the entire volume of the mechanical lithosphere, then faulting will no longer be self-similar. For longer faults, displacement will no longer scale with length as assumed here, resulting in an overestimate of strain in the SP. However, it is possible that a majority of faults in the SP could be thin skinned but contribute minimal bias to the strain estimate here, if shallow faulting is preferentially confined to shorter structures. For example, SP structures shorter than 50 km comprise 60% of the population but only host 22% of the strain (Figure 4b).

5. Conclusions

Spatial variations of areal strain show that the NSP, and the SP in general, have accommodated at least as much strain as the ICP, suggesting that crustal shortening arising from global contraction has substantially deformed all geological units on Mercury. Although structures in the SP generally have less relief and length compared with ICP structures, the increased spatial density of faults in the SP renders the cumulative strain recorded in both types of unit approximately the same. Our results support the view that global contraction continued after the emplacement of the SP, which is consistent with thermal evolution models for Mercury (Tosi et al., 2013) and observations of cross-cutting relations on the surface (Banks et al., 2015). It is therefore likely that the contrast in relief, length, and spatial density observed between the SP and ICP faults reflects structural and/or mechanical differences between the SP and ICP units at the time of fault initiation, rather than a different formation mechanism.

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